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# An implicit discontinuous Galerkin finite element framework for modeling fracture failure of ductile materials undergoing finite plastic deformation

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## Summary

It is a challenge to achieve a complete simulation of fracture failure in ductile materials undergoing large plastic deformation within implicit finite element frameworks due to instability issues. Currently, traditional nodal force or crack surface traction release methods target the direct release of tractions on cracked surfaces within the current time/load step. An abrupt change from a system without cracks to another system with cracks may contribute to the instability issues. Specifically, because of broken meshes, discontinuous Galerkin (DG) methods have an advantage over traditional continuous elements in naturally accommodating crack openings along DG interfaces across elements. To improve the convergence in nonlinear iterations during crack openings, we propose a relaxation scheme for DG formulations to gradually recover the traction-free condition on cracked surfaces. Furthermore, this DG-based relaxation scheme for crack openings in finite plastic media has been consistently formulated within the incomplete interior penalty DG framework. Finally, we have demonstrated a good performance of the proposed implicit DG formulation along with the DG relaxation scheme by successfully solving a nuclear fuel rod structural failure problem with multiple hydride crack openings and the Sandia Fracture Challenge benchmark.

## KEYWORDS

crack opening, ductile fracture, finite plastic deformation, DG relaxation, implicit formulation, traction release

## 1 | INTRODUCTION

Analysis of failure behaviors of ductile materials and structures is critical for the successful design and maintenance of many vital engineering structures, eg, structural failure of nuclear fuel rods in nuclear reactors resulting from hydride cracks induced by hydrogen transport in zirconium alloys. A reliable study on how the failure occurs relies on both the use of appropriate material failure models and high-fidelity computational tools capable of successfully achieving converged solutions and capturing the right physics when cracks open in ductile materials undergoing finite plastic deformation. The current popular numerical techniques in modeling crack opening and propagation problems are based on continuous Galerkin (CG) finite element methods. These techniques include the traditional direct nodal force and surface traction

release methods, the cohesive zone method (CZM), the level set method, the extended finite element method, and the phase field method. The direct nodal force release method is mainly used for modeling crack openings for linear elastic fracture mechanics problems.<sup>1</sup> This method was also adopted in the work of Gao and Kim<sup>2</sup> using an implicit code in modeling void coalescence at finite plastic deformation for multiple voids in a symmetric plane. However, the nodal force release method, in general, has challenges in achieving complete simulations using implicit formulations for multiple and complex void/crack coalescence in media undergoing finite plastic deformation. Crack openings for multiple cracks can be improved by inserting interface elements along with cohesive laws<sup>3-7</sup> in front of crack tips. However, CZM requires the introduction of spurious compliance stiffness into interface elements, which may have negative impacts on the accurate calculation of stresses at crack tips and the opening directions of cracks for multiple-void/crack cases. The level set method<sup>8</sup> has to seek ad hoc methods to compute the traction on crack surfaces because cracked surfaces are not explicitly defined in these methods, which is not accurate. As discussed in the work of Sukumar et al.,<sup>9</sup> the extended finite element method has been successful in modeling fracture problems in which bulk material response is linearly elastic, and they may have instability issues in modeling problems with materials beyond linear elasticity. Finally, similar to the level set method, the phase field method<sup>10</sup> has challenges in accurately computing the stresses around cracks, and research efforts in sharpening crack profiles need further investing. In this work, we propose the discontinuous Galerkin (DG) method, which is a simple physics-based interface framework to improve both the convergence and the accuracy in modeling crack openings in finite plastic media by focusing on implicit computational frameworks along with robust traction release algorithms for crack openings.

DG methods were originally formulated in the works of Baker,<sup>11</sup> Arnold,<sup>12</sup> Douglas and Dupont,<sup>13</sup> Nitsche,<sup>14</sup> and Wheeler,<sup>15</sup> by introducing a jump in primary solutions and an average on fluxes across interfaces among broken elements. As the jump in solutions is small, DG solutions are numerically acceptable in modeling continuous fields. Primary DG methods have several different formulations including symmetric interior penalty Galerkin (SIPG),<sup>16</sup> nonsymmetric interior penalty Galerkin,<sup>17</sup> nonsymmetric Galerkin without penalty,<sup>18</sup> and incomplete interior penalty Galerkin (IIPG).<sup>19</sup> These DG frameworks have been applied to improve or enhance solutions from CG methods for fluid dynamics such as the Navier-Stokes problems,<sup>18-22</sup> solid mechanics,<sup>16,17,23-36</sup> coupled fluid flow and solid mechanics,<sup>37-39</sup> and contact mechanics.<sup>40,41</sup> It should be noted that a full DG discretization for a whole model results in an explosion in unknowns, which is not practical. However, for crack problems, the explosion in unknowns can be avoided by only applying DG elements to areas where cracks may have potential to develop. Specifically, the average in traction and jump in displacement across interfaces in DG formulations were exploited in modeling crack propagation for linear elastic materials,<sup>32,35,42-44</sup> finite strain fragmentation,<sup>45</sup> composite materials,<sup>46</sup> and damaging materials.<sup>47,48</sup> In these formulations, both the generation of newly created crack surfaces and the release of tractions on the cracked surfaces are gracefully handled by DG methods along with cohesive laws. Remarkably, the serious drawback of CZM, ie, the introduction of spurious compliance stiffness, can be circumvented in DG frameworks through adopting extrinsic (rigid) cohesive laws<sup>45</sup> and using zero-thickness CZM elements. It is well known that it is time consuming to insert CZM elements in traditional CG frameworks. The cumbersome construction of zero-thickness CZM elements can be much saved by generating DG interface elements through simply breaking continuous elements,<sup>44,48</sup> which is important with regard to its practical applications.

A few distinctive features of this work are addressed below. First, we have formulated DG methods for crack propagation in finite plastic media in implicit computational frameworks rather than in explicit frameworks. The reliability of explicit and implicit finite element methods has been extensively discussed in the literature.<sup>49-52</sup> In summary, while explicit dynamic simulations are widely used to evaluate high-speed impact problems, quasi-static and low-speed problems are more appropriately and more effectively simulated using implicit dynamic or static solution techniques<sup>49</sup> that are traditionally favored for solid mechanics problems. Furthermore, the fracture failure of ductile materials often involves large plastic deformation. Stresses predicted from explicit formulations may drift away from the yield surface following time integration since they are not guaranteed to be on yield surfaces in each time step. In implicit formulations, predicted stresses are always on yield surfaces by performing stress returning in local material solvers.<sup>53</sup> On the other hand, for microvoid coalescence problems in ductile metals, element sizes are on the levels of  $10^{-7}$  –  $10^{-9}$  m. This would require an extremely large number of time steps for explicit methods to avoid unstable results, which is not practical. These motivate us to adopt an implicit DG framework to accurately and efficiently model crack openings in finite plastic media. Many DG formulations in modeling crack problems have been studied in explicit frameworks.<sup>78</sup> Some implicit DG formulations were adopted for modeling fiber/matrix debonding in the works of Nguyen<sup>44</sup> and Truster and Masud<sup>47</sup> for elastic materials. In the work of Wu et al.,<sup>48</sup> an explicit DG framework for the stress balance equation and an implicit method for the nonlocal damage model were efficiently formulated in modeling fractures in elastic media. In this work, we extend DG methods within an implicit framework to model crack openings in media undergoing finite plastic deformation.

Second, we adopt the traditional direct nodal force/surface traction release methods<sup>54</sup> to enforce the traction-free condition on cracked surfaces within one (current) time/load step. Crack opening criteria are categorized into energy, strain, and stress-based material failure models.<sup>55-59</sup> Indeed, DG methods are able to provide a powerful and flexible computational framework to model interface separation problems with less dependence of material failure models. For example, as reviewed above, crack openings can be simulated by DG methods through hybridizing with softening CZM elements where fracture energy-based traction separation laws or extrinsic cohesive laws are used. By assuming a negligible fracture energy during crack opening events and by following the substepping procedures, the traditional direct nodal force/surface traction release methods can be gracefully implemented in DG frameworks. For crack openings in finite plastic media, the Johnson-Cook failure model,<sup>60-62</sup> ie, the maximum equivalent failure plastic strain, is widely adopted in modeling ductile metal failure problems. In modeling the benchmark of the Sandia Fracture Challenge,<sup>60,63</sup> to capture the sharp dropping behavior in the force-displacement curve, tractions on cracked surfaces are immediately reduced to zero within the current time step through using the element erosion technique within explicit frameworks once the failure plastic strain develops up to the limit. In the work of Gross and Ravi-Chandar,<sup>60</sup> the authors discussed that the damage may occur at much later stages in the deformation where large coalesced voids develop only near the brink of final failure, and thus, fractures progress rapidly. The sudden appearance of damage just prior to failure indicates that the stiffness degradation of the material due to damage can be adequately approximated as a complete loss of stiffness once damage has occurred. More precisely, compared to the dominated plastic dissipation, the remaining critical energy release rate for crack openings is much smaller and is assumed to be negligible, in which a direct opening can be adopted through setting up a critical cumulative damage index up to which the element stiffness is set to zero. For a more detailed discussion on the limitation and improvement of the element erosion method, we refer to the work of Li et al.<sup>64</sup>

In this work, we focus on applying the direct traction release method as well as the Johnson-Cook material failure model for crack openings due to its simplicity and the noninvolvement of additional material laws. Contrary to element erosions, tractions on DG crack interfaces are directly released, and no mass is lost in the proposed DG-based traction release process. It should be underlined that due to the avoidance of substepping procedures required for direct nodal force/surface traction release methods, CZM-based models should be more efficient. For strain- or stress-based crack opening models, DG frameworks hybridized with CZM may be used by constructing a rigid cohesive law using the traction on crack surfaces corresponding to a strain- or stress-based failure criterion and by isolating the critical fracture energy from dissipated plastic energy proposed, eg, in the works of Cotterell et al<sup>65</sup> and Nielsen and Hutchinson.<sup>66</sup> However, we believe that such research effort on this hybrid CZM with DG method for crack openings in ductile materials is out of the scope of this paper and deserves another report.

Third, we propose a novel DG relaxation scheme to greatly improve on the instability issues in nonlinear Newton-Raphson iterations during crack opening in finite plasticity media. Unlike explicit methods, implicit methods have challenges in achieving converged solutions for many highly nonlinear problems. While DG methods<sup>27,43-45,47,48</sup> are able to provide graceful formulations for crack problems, their fully and consistent implicit formulations most likely exhibit similar instability issues to the traditional CG methods during Newton-Raphson iterations especially for crack openings in finite plastic media. In direct nodal force release or traction release methods, if a few interfaces across elements are identified to separate each other according to converged solutions in the current time/load step, the tractions on those interfaces will be released to zero within the current time/load step through solving an additional substep unloading problem to gradually reduce nodal forces before going to the next time/load step. More precisely, in those direct traction release methods, tractions on cracked surfaces are treated as the prescribed traction boundaries in the separation process of interfaces. It is crucial to note that unlike explicit methods, the performance of the Newton iterations in implicit formulations is sensitive to the behavior change of two nonlinear system equations, one for a problem without cracks and the other with newly introduced and highly localized traction boundaries due to the nucleation of cracks, which may eventually lead to diverged solutions especially for crack openings in media where materials yield and stress states are very complex. The idea of our proposed DG relaxation scheme is to avoid such a sudden change in nonlinear systems in DG formulations by incorporating a parameter in front of DG interface integrals and by gradually relaxing this parameter to zero when a crack opens.

Finally, we adopt the IIPG method and derive a complete consistent formulation on this method along with the proposed DG relaxation scheme for crack openings in finite plastic media. As discussed in the works of Liu et al,<sup>29,39,67</sup> there are some issues for symmetric DG or SIPG formulations for solid mechanics problems. The first issue is that SIPG is sensitive to the selection of the stabilized parameter. The second issue is that for nonlinear problems, SIPG has difficulty in computing the residual contributed from the added interface variational part in terms of the inner product of the average on stress and the jump on solution (displacement). The second issue on the residual computation for symmetric DG methods has

been solved in the works of Ten Eyck et al,<sup>26</sup> Noels and Radovitzky,<sup>33</sup> and Truster et al,<sup>34</sup> by a modification of the SIPG method. The ideas of these modifications are to introduce the average on a stress-like variable rather than using the exact Cauchy stress in the symmetric numerical flux term. These stress-like variables are written in a format of the gradient of displacement and the material or acoustic tensor. Consequently, with these modifications, consistent DG formulations are possible, eg, in the works of Truster et al<sup>34</sup> and Truster and Masud.<sup>47</sup> However, the symmetric property of tangent stiffness matrices is not preserved if complete consistent formulations are followed in these modifications for finite deformation problems. As discussed in the work of Truster et al,<sup>34</sup> however, the linearization in the modified formulations involves seeking the derivative for “acoustic tensor” that is a function of deformation, which is complex and difficult to obtain for some problems with complex material behaviors. In addition, the quality of tangent stiffness may be degraded because it requires computing for the third-order derivative of potential functions, which is equivalent to taking the derivative on stresses or strains with respect to displacements. Furthermore, the performance of the modified symmetric DG formations in the works of Truster et al<sup>34</sup> and Truster and Masud<sup>47</sup> needs to be addressed for modeling finite plasticity problems. On the other hand, the IIPG method is a proved high-fidelity DG formulation<sup>29,30,67</sup> for nonlinear solid mechanics problems. Complete consistent formulations can be achieved for the IIPG method for finite plasticity problems because it does not involve the numerical flux term and only needs the penalty term. The high robustness of IIPG formulations has been demonstrated in the work of Liu et al,<sup>30</sup> where plastic strains are up to several hundred percent and the penalty parameter is simply set up to 1 for all elements, among which element sizes and material moduli have differences in several orders of magnitude. Based on the above discussions, we adopt the IIPG method in this work for modeling crack openings in finite plasticity media.

The remaining sections of this paper are organized as follows. In Section 2, we present a few important nonlinear continuum mechanics terminologies, and some material failure models are also briefly reviewed. Crack opening problems in finite plasticity media are mathematically defined in Section 3. Consistent and objective IIPG formulations for both direct traction release and DG relaxation are derived in Section 4. Specifically, the objectivity of a traction release algorithm for crack openings in finite deformation media is discussed. Tangent stiffness matrix and residual force are summarized in Section 5. A diagram on nonlinear solution procedures is also documented in this section. In Section 6, we demonstrate a good performance of the proposed implicit DG formulation along with the DG relaxation scheme by successfully solving a nuclear fuel rod structural failure problem with multiple hydride crack openings and the Sandia Fracture Challenge benchmark. Conclusions are summarized in Section 7.

## 2 | NONLINEAR CONTINUUM MECHANICS

In this section, we briefly review a few important nonlinear continuum mechanics terminologies to leverage the mathematical statements on crack opening problems in finite strain media discussed in later sections. As finite plasticity involves, we discuss the decompositions of deformation gradients in the first subsection of this section. In the second subsection, stress and strain measurements are discussed. In the third subsection, we summarize material laws and the  $J_2$  plasticity theory in terms of finite deformation. Finally, we summarize some popular material failure models for the debonding of inclusion/bulk interfaces and the fracture onset and propagation of ductile materials.

### 2.1 | Multiplicative decomposition

Hypoelastoplasticity and hyperelastoplasticity<sup>68</sup> are two major theories for finite plasticity problems. We will focus on hyperelastoplasticity in this section. As the magnitude of plastic strain often goes up to several hundred percent for material failure problems at microscales, we adopt the multiplicative decomposition theory. Let  $B_t \in R^3$  be the current deformed configuration of a continuum body and  $\partial B_t$  be the boundary of  $B_t$ .  $B_t$  migrates from its reference configuration  $B_0 \in R^3$  bounded with  $\partial B_0$ . Furthermore, we decompose  $\partial B_t$  into  $\partial B_t^p$  and  $\partial B_t^c$ . More precisely,  $\partial B_t^p$  represents the pre-described boundary evolved from  $\partial B_0$ .  $\partial B_t^c$  indicates the set of new boundary surfaces created by crack openings in the deformed body  $B_t$ , which is a part of the problem solution. A particle  $X \in B_0$  is mapped into  $x \in B_t$  through a mapping function as follows:

$$x(X, t) = \phi(X, t), \quad X \in B_0, x \in B_t, t \in (0, +\infty). \quad (1)$$

The deformation gradient  $F$  is defined as the partial derivative of the mapping function with respect to  $X$  as follows:

$$F(X, t) = \frac{\partial \phi(X, t)}{\partial X}, \quad F \in \text{Lin}, \quad (2)$$

where  $\text{Lin}$  defines the tensor space, a set of all linear mappings of a finite-dimensional vector space into itself. In multiplicative decomposition theory,  $F$  is decomposed into a plastic deformation gradient tensor  $F^p(X, t) \in \text{Lin}$  and an elastic deformation  $F^e(\phi^{-1}(X, t))$  as follows:

$$F(X, t) = F^e(\phi^{-1}(X, t)) F^p(X, t), \quad F^e, F^p \in \text{Lin}, \quad (3)$$

where  $x \in B_t$ , which is the intermediate configuration or fictitious domain. Material laws defined in this fictitious domain always satisfy the principle of material frame indifference.

## 2.2 | Strain and stress measurements

Following the definition of the deformation gradient in the previous subsection, we continue to define three important stress and strain conjugate pairs that are useful for developing governing equations for finite deformation problems in different configurations. These three conjugate pairs are  $(\sigma, e)$ ,  $(P, F)$ , and  $(S, E)$ , where  $\sigma$  is the Cauchy stress,  $P$  is the first Piola-Kirchhoff stress, and  $S$  is the second Piola-Kirchhoff stress. Corresponding to stress tensors  $\sigma$ ,  $P$ , and  $S$ ,  $e$  is the Almansi strain tensor,  $F$  is the deformation gradient tensor, and  $E$  is the Green-Lagrange strain tensor, respectively. The Almansi and Green-Lagrange strain tensors are further defined as follows:

$$e = \frac{1}{2}(I - b^{-1}), \quad e, b \in \text{Lin}; \quad E = \frac{1}{2}(C - I), \quad E, C \in \text{Lin}, \quad (4)$$

where  $C$  and  $b$  are the right Cauchy-Green strain tensor and the left Cauchy-Green tensor, respectively. They can be obtained by using the deformation gradient tensor  $F$  as follows:

$$C = F^T F, \quad C, F \in \text{Lin}; \quad b = F F^T, \quad b, F \in \text{Lin}. \quad (5)$$

The Cauchy stress, the first Piola-Kirchhoff stress, and the second Piola-Kirchhoff stress are frequently transferred to each other through the following relations:

$$\sigma = F S F^T / J, \quad P = J \sigma F^{-T}, \quad (6)$$

where  $J$  is the determinant of the deformation gradient tensor  $F$ . For materials in plastic states, the elastic deformation gradient  $F^e$  obtained from the multiplicative decomposition of total deformation gradient tensor  $F$  should be used for stress transformations in Equation (6).

## 2.3 | Plasticity material laws

For hyperelasticity, stresses can be directly obtained from elastic potentials based on the Helmholtz free energy functions. As the multiplicative decomposition on the deformation gradient is adopted, elastic potentials can be formulated in terms of the left Cauchy-Green tensor  $C^e$  in the intermediate configuration. The second Piola-Kirchhoff stress  $S$  and the elasticity tensor  $D^e$  in the intermediate configuration can be derived as

$$S = 2 \frac{\partial \bar{W}}{\partial C^e}, \quad C^e, S \in \text{Lin}; \quad D^e = 4 \frac{\partial^2 \bar{W}}{\partial C^{e2}}, \quad C^e \in \text{Lin}, \quad D^e \in \text{Lin} \times \text{Lin}, \quad (7)$$

where  $C^e = F^{eT} F^e$ . For plastic response, the total plastic strain cannot be directly computed from potentials and must be obtained through integrating the rate of change of the plastic strain. The plastic velocity gradient  $L^p$  serves as such rate of change of the plastic strain, and it is defined as follows:

$$L^p = \dot{F}^p F^{p-1} = L^p \in \text{Lin}. \quad (8)$$

Yield, plastic flow, and hardening functions are required in plasticity models. These functions can be derived in terms of the second Piola-Kirchhoff stress  $S$  and hardening variables  $Q$  defined in the intermediate configuration as follows:

$$\begin{cases} \lambda Y(S, Q) = 0, & \lambda \in [0, +\infty), Y \in R & (\text{Yield Function}) \\ L^p = \lambda \frac{\partial G(S, Q)}{\partial S}, & G \in R & (\text{Flow Potential}) \\ \dot{Q} = -\lambda H(S, Q), & H \in R, Q \in R^m. & (\text{Hardening Function}) \end{cases} \quad (9)$$

## 2.4 | Criteria for material failure

Many material failure theories for the determination of crack nucleation and propagation are available in the literature. As this work focuses on numerical methods, however, only some important failure criteria that are more appropriate for ductile materials will be discussed here. Fracture criteria for defect/bulk matrix interface debonding are categorized into fracture energy (eg, cohesive laws), plastic strain, and stress-based crack opening theories.<sup>55–58</sup> Among these criteria, the local stress criterion is popular for evaluating the onset of inclusion/matrix interface debonding. It is based on the equivalent interface stress  $\bar{\sigma}_n$  between the inclusions and the bulk matrix and can be given as follows<sup>57,58</sup>:

$$\bar{\sigma}_n = \sigma_n + \gamma \tau_s > \sigma_0, \quad (10)$$

where  $\gamma$  is the weighting factor,  $n$  is the normal vector of the surface,  $\sigma_n = (\sigma n) \cdot n$  is the normal component of the traction,  $\tau_s = |\sigma n - \sigma_n n|$  is the tangent component of the traction, and  $\sigma_0$  is the interface failure strength. As illustrated in Figure 8, for void coalescence in ductile metals, the ligaments associated with growing voids/cracks undergo serious necking and large plastic strain. Rather than using local stress criteria for crack initiation on inclusion/bulk matrix interfaces, the critical wall thickness between two necked ligaments or the local failure plastic strain developed in ligaments is more appropriate for the final tearing-off of necked ligaments, as discussed in the works of Thomason,<sup>59</sup> Morin et al,<sup>69</sup> Pardoen and Hutchinson,<sup>70,71</sup> and Tvergaard and Hutchinson.<sup>72</sup> The Johnson-Cook failure model<sup>60–62</sup> is widely adopted for modeling ductile metal failure problems. This model uses the maximum equivalent plastic strain as the failure index that is a function of stress triaxiality or average triaxiality over elements. One of such failure strain functions can be described as follows:

$$\epsilon^f = D_1 + D_2 e^{-D_3 \sigma^*}, \quad (11)$$

where  $\sigma^*$  is the triaxiality parameter defined by the ratio of the mean stress and von Mises stress. More complex failure models related to the Lode angle and shear strain can be found in the works of Boyce et al<sup>63</sup> and Xue.<sup>73</sup>

## 3 | CRACK OPENING PROBLEMS

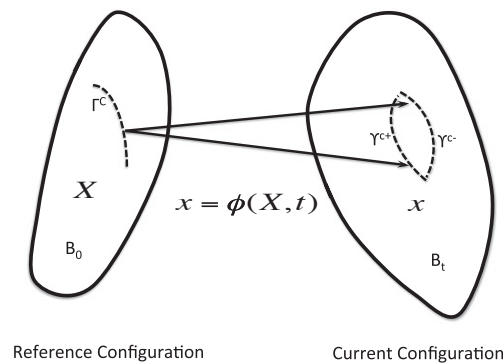
In this section, we mathematically define problems with crack openings in finite strain media. We first discuss the physics and mathematics aspects of crack problems in the first subsection. Following the discussion on the crack opening physics, we make two mathematical statements, namely, *ProblemA* and *ProblemB*, which enable us to have a mathematically well-defined crack opening problem in the framework of finite deformation.

### 3.1 | Crack opening

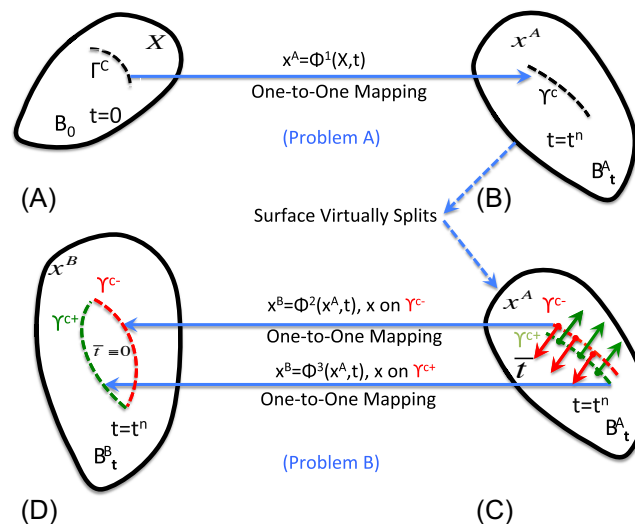
Let  $\Gamma^C$  be an interior surface set defined in the reference configuration and shown in Figure 1.  $\Gamma^C$  is a virtual surface set that has potential for crack openings.  $\Gamma^C$  is eventually mapped into two surfaces  $\gamma^{c+}$  and  $\gamma^{c-}$  in the current configuration after a crack opens. Tractions on these two new surfaces in the current configuration  $B_t$  must be free for mode I-like cracks, ie,

$$\bar{t}(x, t) = \sigma(x, t) \mathbf{n}(\mathbf{x}, \mathbf{t}) = 0, \quad x \in \gamma^{c+} \cup \gamma^{c-}, \quad t \in (0, +\infty), \quad \sigma \in \text{Lin}, \quad \bar{t} \in R^3. \quad (12)$$

Obviously, the mapping  $\phi^1$  in Figure 1 for the potential crack surface on  $\Gamma^C$  from reference configuration  $B_0$  to the corresponding two cracked surfaces on  $\gamma^{c-}$  and  $\gamma^{c+}$  in current configuration  $B_t$  is not a one-to-one mapping. This implies that crack opening problems would not be well defined unless we make some additional assumptions on how cracks open. Practically, this difficulty for crack problems can be avoided by solving two problems, *ProblemA* and *ProblemB*, which will be defined below.



**FIGURE 1** Mapping on the crack surface: non-one-to-one mapping



**FIGURE 2** Mapping on the crack surface. A, potential crack surface in the reference configuration; B, deformed potential crack surface in the current configuration; C, surface virtually opens without traction release (two identical cracked surfaces should coincide each other but are technically shifted); D, traction-free condition on cracked surfaces are enforced. Note: The superscripts “A” and “B” over  $B_t$  indicate two different current configurations (but within the same time step) for Problem A and Problem B, Respectively [Colour figure can be viewed at [wileyonlinelibrary.com](http://wileyonlinelibrary.com)]

We assume that the surface  $\Gamma^c$  in the reference configuration shown in Figure 2A is first mapped into a surface  $\gamma^c$  in the current configuration without crack opening shown in Figure 2B. Because of no crack opening in this stage, the mapping  $\phi^1$  is a one-to-one mapping. For this stage, the problem, named as *ProblemA*, can be solved by applying well-defined boundary conditions (prescribed) and material constitutive laws. After *ProblemA* is solved and its solution converges, we check if the crack opening criteria are satisfied for this virtual surface according to the solutions of *ProblemA*. If satisfied, we define a new problem called *ProblemB* in the same current configuration (time is not changing) through (i) splitting surface  $\gamma^c$  into two identical surfaces  $\gamma^{c+}$  and  $\gamma^{c-}$ ; (ii) applying the traction solution obtained from *ProblemA* on the surface  $\gamma^c$  to the two cracked surfaces  $\gamma^{c+}$  and  $\gamma^{c-}$  as new initial traction boundary conditions of *ProblemB*, that is, more precisely, the prescribed traction boundary condition or the Neumann boundary condition is applied on cracked surfaces; (iii) setting up the stress and strain solutions from *ProblemA* as initial stresses and strains for *ProblemB*; and (iv) enforcing the traction-free condition on both  $\gamma^{c+}$  and  $\gamma^{c-}$ . This involves substep unloading procedures to gradually reduce traction to zero. We argue that mappings  $\phi^2$  and  $\phi^3$  from Figure 2C to Figure 2D are one-to-one mappings, and therefore, *ProblemB* is also a well-defined problem because cracked surfaces become prescribed traction boundaries. We underline that crack opening problems become mathematically well-defined problems by following procedures (i)-(iv) discussed above. For important crack opening criteria and traction release methods, we cover them in later sections.

### 3.2 | Problem statements

With the previous review and discussion on stress and strain measurements, material laws, and crack openings, we make two statements for crack opening problems in finite elastoplastic media as follows.

**ProblemA** (Hyperelastoplasticity without crack openings) Given a set of virtual surfaces  $\Gamma^C$  in the reference configuration, find  $(u^A(x, t), \sigma^A(x, t), F^{Ap}(X, t))$  that satisfy the equations below:

$$\begin{cases} \nabla \cdot \sigma^A + \mathbf{b} = 0, & (\text{Stress Equilibrium}) \\ S^A = 2 \frac{\partial W(F^{Ae})}{\partial C^{Ae}}, & (\text{Hyperelastic Potential}) \\ \lambda^{Ae} Y(S^A, Q^A) = 0, & (\text{Yield Condition}) \\ L^{pA} = \lambda^A \frac{\partial G(S^A, Q^A)}{\partial C^{Ae}}, & (\text{Flow Potential}) \\ \dot{Q}^A = -\lambda^A H(S^A, Q^A), & (\text{Hardening Rule}) \end{cases} \quad (13)$$

with the prescribed displacement on boundary  $\gamma^u$  and traction on boundary  $\gamma^t$

$$\begin{cases} u^A(x) = \bar{u}(x), & x \in \gamma^u \\ \sigma^A n = \bar{t}(x), & x \in \gamma^t \end{cases} \quad (14)$$

and the complementary relation for material yielding

$$\begin{cases} \lambda^A = 0, & \text{if } Y < 0 \\ \lambda^A > 0, & \text{if } Y = 0, \end{cases} \quad (15)$$

where  $\mathbf{b}$  is the body force. The measurement variables with superscript  $A$  simply indicate that they are defined for *ProblemA*, by which potential confusions on notations will be avoided for going through the statement of *ProblemB* below.

After *ProblemA* is solved, the interface failure criteria will be checked for each individual interface in the set of interior surfaces. If the failure criteria are satisfied for an interior surface, this interior surface will be split into a pair of cracked surfaces, and the traction obtained from *ProblemA* will be applied to these two cracked surfaces as external surface traction loadings for *ProblemB*. More precisely, the cracked surfaces are now categorized as a new part of the boundary of the problem. The interior set  $S$  will be updated by excluding the cracked surface set  $\gamma^c$  due to our traditional CG implementation on boundary conditions. If the failure criteria are not satisfied for all interior surfaces, *ProblemB* need not be solved as no crack opens and the solution will be going to the next time step. In addition, we will adopt local stress/plastic strain failure criteria to determine when cracks open. The calculation on local stresses/strains on interfaces is based on DG formulations, in which tractions on surfaces are obtained by incorporating material constitutive laws of bulk elements that the surface sits on. Unlike CZM that adopts softening constitutive laws for interface traction calculations for modeling crack openings, the proposed DG implementation in this study is to directly release tractions on interfaces, and thus, there is no need to provide an additional constitutive law that is softening and different from bulk materials for interface traction calculation.

**ProblemB** (Hyperelastoplasticity with crack opening) Given a set of cracked surfaces  $\gamma^{c+}$  and  $\gamma^{c-}$  in the current configuration, find  $(u(x, t), \sigma(x, t), F^p(X, t))$  at current time  $t^n$  that satisfy the equations below:

$$\begin{cases} \nabla \cdot \sigma + \mathbf{b} = 0, & (\text{Stress Equilibrium}) \\ S = 2 \frac{\partial W(F^e)}{\partial C^e}, & (\text{Hyperelastic Potential}) \\ \lambda Y(S, Q) = 0, & (\text{Yield Condition}) \\ L^p = \lambda \frac{\partial G(S, Q)}{\partial C^e}, & (\text{Flow Potential}) \\ \dot{Q} = -\lambda H(S, Q), & (\text{Hardening Rule}) \end{cases} \quad (16)$$

with (i) the prescribed displacement and traction boundary conditions:

$$\begin{cases} u(x) = \bar{u}(x), & x \in \gamma^u \\ \sigma(x)n = \bar{t}(x), & x \in \gamma^t; \end{cases} \quad (17)$$

(ii) the initial stress, strain, traction (for newly cracked surfaces), and state variable conditions obtained from *ProblemA*:

$$\begin{cases} \sigma(x) = \sigma^A(x), & x \in B_t \\ F^p(\phi^{-1}(x)) = F^{pA}(\phi^{-1}(x)), & x \in B_t \\ Q^p(\phi^{-1}(x)) = Q^{pA}(\phi^{-1}(x)), & x \in B_t \\ \sigma(x)n(x) = \sigma(x)^A n(x), & x \in \gamma^{c+} \\ \sigma(x)n(x) = \sigma(x)^A n(x), & x \in \gamma^{c-}; \end{cases} \quad (18)$$

(iii) the required traction-free conditions on the cracked surfaces:

$$\begin{cases} \sigma(x)n = 0, & x \in \gamma^{c+} \\ \sigma(x)n = 0, & x \in \gamma^{c-}; \end{cases} \quad (19)$$

and (iv) the complimentary material yield condition:

$$\begin{cases} \lambda = 0, & \text{if } Y < 0 \\ \lambda > 0, & \text{if } Y = 0. \end{cases} \quad (20)$$

The main difference between the two problem statements is that additional prescribed traction boundary conditions are enforced on newly opened surfaces for Problem B. The initial conditions for *ProblemB* are based on the converged solution from *ProblemA*. The initial tractions on cracked surfaces must be reduced to zero by employing some substepping unloading techniques. While only the hyperelastoplasticity framework has been briefly reviewed in this work, our DG formulations have been derived, implemented, and evaluated for finite plasticity problems with regard to both hyperelastoplasticity and hypoelastoplasticity models. For hypoelastoplasticity models and their corresponding computational corotational framework, we refer to the work of Hughes and Pister.<sup>53</sup>

## 4 | IIPG FORMULATION ON PROBLEM A

In this section, we introduce some key notations on DG methods. Instead of presenting a thorough derivation, we briefly summarize the results of the IIPG formulation for *ProblemA* defined in the previous section as a detailed formulation within the updated Lagrange framework has been documented in the work of Liu et al<sup>30</sup> for finite plasticity problems without cracks.

Let a continuum body  $B_t$  be discretized into a broken set

$$\chi = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}. \quad (21)$$

The set for the pair of interior interfaces across elements is denoted by

$$\mathbf{S} = \{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_m\}. \quad (22)$$

As a consequence of the discretization of  $B_t$ ,  $\partial B_t$ , the boundary of  $B_t$ , is discretized into prescribed surface elements.  $\partial B_t$  is further partitioned into  $\gamma^u$  and  $\gamma^t$  for the prescribed displacement and traction boundary surfaces, respectively. Boundary conditions on  $\partial B_t$  could be weakly enforced by DG methods, ie, the formulation of Burman.<sup>74</sup> However, in this study, we only apply DG methods on interior element faces. Boundary conditions are strongly enforced by following traditional CG implementations.

For two elements sharing an interior interface, we name one of the two elements as the right element and the other as the left element. We define

$$[w] = w^L - w^R \quad (23)$$

and

$$\{w\} = \frac{1}{2}(w^L + w^R) \quad (24)$$

as the jump and averaged variables across the broken element interfaces, respectively. Following the notations defined in Equations (21)-(24), the IIPG formulation in terms of the residual form for finite strain problems is summarized below:

$$\begin{aligned}
 R(u, \delta u) = & \sum_{e \in \chi} \int_e \sigma(u) : \nabla(\delta u) dv - \sum_{\partial e \in S} \int_{\partial e} \{\sigma n\} \cdot [\delta u] ds \\
 & + \sum_{\partial e \in S} \delta_p \frac{G_0}{h_0} \int_{\partial e} [u] \cdot [\delta u] ds - \sum_{\partial e \in \gamma^t} \int_{\partial e} \bar{t} \cdot \delta u ds - \sum_{e \in \chi} \int_e \mathbf{b} \cdot \delta u dv,
 \end{aligned} \tag{25}$$

where  $\delta_p$ ,  $G_0$ , and  $h_0$  are the penalty or stabilized parameter, material shear modulus, and the square root of the interface face area.

Next, we need to consistently linearize Equation (25). A detailed derivation was documented in the work of Liu et al.<sup>29</sup> In the aforementioned work,<sup>29</sup> the linearization on Equation (25) is based on taking the material time derivative of variables. The linearized equations are eventually formulated in terms of the spatial velocity gradients. Final linearized results in terms of the incremental displacement are obtained through applying the backward Euler integration in the time domain. We summarize the linearized equations below:

$$\begin{aligned}
 R(\Delta u, \delta u) = & \sum_{e \in \chi} \int_e (\mathbf{grad}(\Delta u) : c : \mathbf{grad}(\delta u) + \mathbf{grad}(\Delta u) \sigma : \mathbf{grad}(\delta u)) dv \\
 & - \sum_{\partial e \in S} \int_{\partial e} \{c : \mathbf{grad}(\Delta u) + \mathbf{grad}(\Delta u) \sigma\} n \cdot [\delta u] ds \\
 & + \sum_{\partial e \in S} \frac{\delta_p G_0}{h_0} \int_{\partial e} ([\Delta u] \cdot [\delta u] + (g^{\Delta u} n) \cdot n [\delta u]) ds - \sum_{\partial e \in \gamma^t} \int_{\partial e} (\dot{\bar{t}} \cdot \delta u + (g^{\Delta u} n) \cdot n \bar{t} \cdot \delta u) ds,
 \end{aligned} \tag{26}$$

where  $\Delta u$  is the incremental displacement and  $\mathbf{grad} \equiv \nabla$ . The variable  $g^{\Delta u}$  in Equation (26) is defined as follows:

$$g^{\Delta u} = \mathbf{trace}(\mathbf{grad}(\Delta u)) I - \mathbf{grad}^T \Delta u, \tag{27}$$

where  $I$  is the second-order identity tensor, and the superscript  $T$  is the transpose of the second-order tensor.

## 5 | IIPG FORMULATION ON PROBLEM B

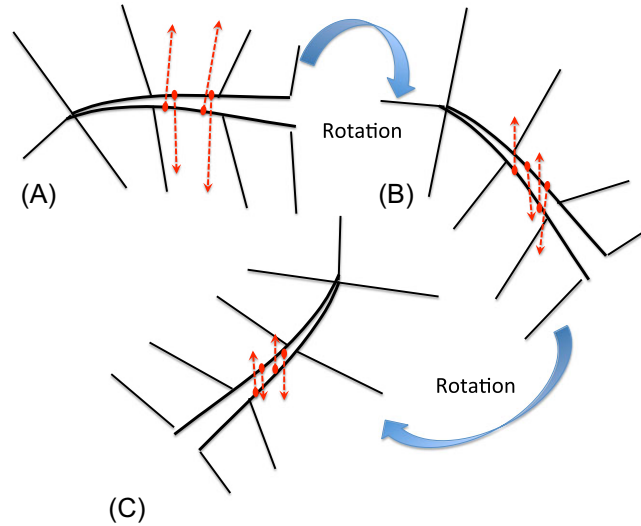
### 5.1 | Traction-free condition on cracked surfaces

The essence of the crack opening problem defined in *Problem B* in the previous section is a complete enforcement of the traction-free condition on cracked surfaces after cracks open. In this work, two schemes are studied and compared to enforce the traction-free conditions on cracked surfaces. The first scheme we will focus on in this section is the direct traction release method, which is a standard crack opening scheme for modeling crack problems. The second scheme proposed in this study is based on a DG relaxation method, which we will discuss in the next section.

### 5.2 | Direct traction release method

#### 5.2.1 | Objective traction release algorithm

For the direct traction release method, the interior of a continuum body  $B_t$  is now treated as additional and newly prescribed traction boundaries. The traction-free condition must be enforced on the newly created crack surfaces. However, a direct traction release in one step, in general, results in nonconvergent solutions for highly nonlinear problems especially undergoing large plastic deformation.<sup>54</sup> More precisely, the tractions on the cracked surfaces must be gradually reduced to zero, which requires unloading substep techniques. However, we only know the cracked surface tractions including their magnitude and direction at the first unloading substep, which is at its stage. This is very different from small deformation problems. In small deformation problems, the directions of cracked surface tractions can be fixed for all subsequent substeps. However, for finite deformation problems, the fixed tractions during unloading, in general, result in nonphysical issues. As shown in Figure 3, for example, the initial tensile traction on the cracked surface is assumed to be along the normal direction of the surface. A potential large rotation of the body may occur in Figure 3 during the traction release process. If we fix the traction direction during the unloading, surface shear and compressive loadings would be present



**FIGURE 3** Fixed traction direction during traction release (Nonphysical issues). A, initial normal traction on cracked surfaces; B, traction reduction and structure undergoing large rotation motion (Traction in shear occurs); C, Traction Reduction and Further Rotation (Traction turns to be compressive loading) [Colour figure can be viewed at [wileyonlinelibrary.com](http://wileyonlinelibrary.com)]

in subsequent unloading substeps, which are not realistic. This reminds us of the importance of correctly prescribing the traction change during the traction release for finite deformation problems.

To avoid nonphysical issues, we propose an objective traction release algorithm (OTRA) to enforce the traction-free condition on crack surfaces. As shown in Figure 4, a pair of surfaces  $\gamma^{c+}$  and  $\gamma^{c-}$  in the current configuration is virtually split from the same crack surface  $\gamma^c$ . We focus only on the surface on the  $\gamma^{c+}$  side in the following discussion. Traction  $\bar{t}_0^{c+}$  on  $\gamma^{c+}$  is the initial value obtained from *Problem A*. Our idea is to pull back  $\bar{t}_0^{c+}$  into the reference configuration (undeformed state) as follows:

$$\bar{T}_0 = (F_0^n)^{-1} \bar{t}_0^{c+}, \quad (28)$$

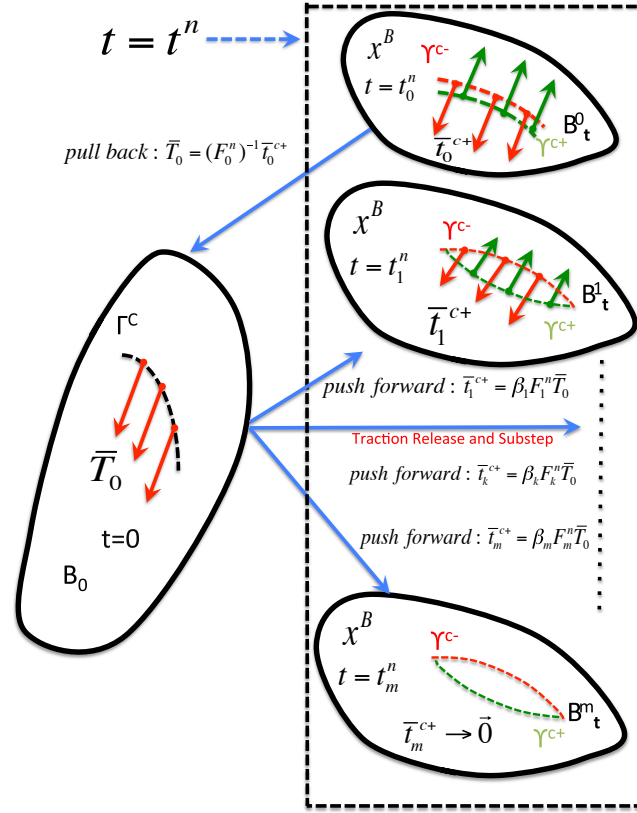
where  $F_0^n$  is the deformation gradient on  $\gamma^{c+}$  at time  $t^n$  before releasing tractions (substep 0). We now assume that  $\bar{T}_0$  is always fixed in its direction in the reference configuration during the traction release. Because of the spatial formulation we adopt in this work, we need to prescribe the traction (treated as an external load) change on the cracked surfaces in the current configuration during the traction release. This can be easily done through pushing  $\bar{T}_0$  forward to the current configuration by using current substep deformation gradient  $F_k^n$  as follows:

$$\bar{t}_k^{c+} = \beta_k F_k^n \bar{T}_0, \quad (29)$$

where  $\beta_k$  is the load reduction factor for the  $k$ th unloading substep in the current load step  $n$ . Instead of using  $\beta_k \bar{t}_0^{c+}$ , we apply  $\bar{t}_k^{c+}$  computed in Equation (29) to implement the traction release for the cracked surface  $\gamma^{c+}$  in the current configuration. More precisely, the traction  $\bar{t}^{c+}$  is no longer fixed in the spatial formulation but depends on the unknown deformation gradient  $F$ . The traction  $\bar{t}^{c+}$  behaves as a follower load. It can be easily verified that no shear force occurs on the cracked surfaces if only rigid body motion and only normal tractions are present before the traction release process starts. Because of this feature of the proposed traction release procedure, we call it the OTRA.

### 5.2.2 | Consistent IIPG formulation

In this section, we derive a consistent formulation for crack openings in media with finite deformation. In this formulation, we focus on releasing tractions on cracked surfaces within the current time step. For DG formulations hybridized with CZM methods or cohesive laws for crack openings, we refer to the works of Molari et al.,<sup>32</sup> Hansbo and Salomonsson,<sup>42</sup> Mergheim et al.,<sup>43</sup> Nguyen,<sup>44</sup> Radovitzky et al.,<sup>45</sup> Truster and Masud,<sup>47</sup> and Wu et al.<sup>48</sup>



**FIGURE 4** Objective traction release algorithm: initial tractions on cracked surface are pulled back to the reference configuration. the traction in the reference configuration is fixed in direction. the fixed traction in the reference configuration is pushed forward to the current configuration using deformation gradient  $F_k^n$  (at Time  $t = t^n$  and Substep  $k$ );  $B_t^m$  indicates the current configuration of  $B_t$  at the  $m$ th unloading substep [Colour figure can be viewed at [wileyonlinelibrary.com](http://wileyonlinelibrary.com)]

The variational formulation in terms of the residual form is given below:

$$\begin{aligned}
 R(u, \delta u) = & \sum_{e \in \mathcal{X}} \int_e \sigma(u) : \nabla(\delta u) dv - \sum_{\partial e \in S/\gamma^c} \int_{\partial e} \{\sigma n\} \cdot [\delta u] ds + \sum_{\partial e \in S/\gamma^c} \frac{\delta_p G_0}{h_0} \int_{\partial e} [u] \cdot [\delta u] ds \\
 & - \sum_{\partial e \in \gamma^+} \int_{\partial e} \bar{t} \cdot \delta u ds - \sum_{e \in \mathcal{X}} \int_E \mathbf{b} \cdot \delta u dv - \sum_{\partial e \in \gamma^{c+}} \int_{\partial e} \bar{t}^{c+} \cdot \delta u ds - \sum_{\partial e \in \gamma^{c-}} \int_{\partial e} \bar{t}^{c-} \cdot \delta u ds,
 \end{aligned} \quad (30)$$

where the last two terms on the right-hand side are the contribution from the crack on surface  $\gamma^c$ , which is virtually split into two identical surfaces  $\gamma^{c+}$  and  $\gamma^{c-}$  shown in Figure 4. Comparing Equation (38) with Equation (25) for finite plasticity without crack openings, only these two prescribed traction boundaries for newly created cracks are present in Equation (38). We only need to linearize these two additional traction-related terms in Equation (38) since all other terms have been linearized in Equation (26). Moreover, for derivation, only one side of surfaces, ie,  $\gamma^{c+}$  in Equation (38), is focused on. The variational term for the prescribed traction on the surface ( $\gamma^{c+}$ ) of the pair of the cracked surfaces is pulled out from Equation (38) as follows:

$$R_{c^+}(u, \delta u) = \sum_{\partial e \in \gamma^{c+}} \int_{\partial e} \bar{t}^{c+} \cdot \delta u ds. \quad (31)$$

In order to linearize Equation (31), we take the material time derivative for each variable in the equation as follows:

$$\dot{R}_{c^+}(u, \delta u) = \sum_{\partial e \in \gamma^{c+}} \int_{\partial e} \left( \dot{\bar{t}}^{c+} \cdot \delta u ds + \bar{t}^{c+} \cdot \dot{\delta u} ds \right), \quad (32)$$

where the notation  $\dot{\cdot}$  over heads of variables indicates the operation of the material time derivative. The material time derivative of variable  $F$  is given below:

$$\dot{F} = lF, \quad (33)$$

where  $l$  is the velocity gradient. The material time derivative for  $ds$  is derived in the work of Liu et al<sup>30</sup> as follows:

$$\dot{ds} = l_s : n \otimes nds, \quad (34)$$

where  $n$  is the normal vector of the crack surface.  $l_s$  in Equation (34) is the Lamb surface velocity gradient and can be expanded as

$$l_s = \text{trace}(l)I - l^T. \quad (35)$$

The material time derivative of traction  $\bar{t}^{c+}$  is derived by using Equations (33) and (29) as follows:

$$\dot{\bar{t}}^{c+} = \beta \left( \dot{F}\bar{T}_0 + F\dot{\bar{T}}_0 \right) = \beta l F \bar{T}_0 = \bar{l}^{c+}, \quad (36)$$

where the subscript  $k$  and superscript  $n$  in Equation (29) are dropped for a concise derivation. Inserting Equation (36) and Equation (34) into Equation (32), we have the linearized virtual work for the cracked surface  $\gamma^{c+}$  as follows:

$$\dot{R}_{c+}(u, \delta u) = \sum_{\partial e \in \gamma^{c+}} \int_{\partial e} (l + l_s : n \otimes nI) \bar{t}^{c+} \cdot \delta u ds. \quad (37)$$

Adding Equation (37) into Equation (26) and applying the backward Euler time integration scheme, finally, we obtain the linearized virtual work for finite deformation problems with crack openings as follows:

$$\begin{aligned} R_c(\Delta u, \delta u) = & \sum_{e \in \chi} \int_e (\mathbf{grad}(\Delta u) : c : \mathbf{grad}(\delta u) + \mathbf{grad}(\Delta u) \sigma : \mathbf{grad}(\delta u)) dv \\ & - \sum_{\partial e \in S/\gamma^c} \int_{\partial e} \{c : \mathbf{grad}(\Delta u) + \mathbf{grad}(\Delta u) \sigma\} n \cdot [\delta u] ds + \sum_{\partial e \in S/\gamma^c} \frac{\delta_p G_0}{h_0} \int_{\partial e} ([\Delta u] \cdot [\delta u] + (g^{\Delta u} n) \cdot n[u] \cdot [\delta u]) ds \\ & - \sum_{\partial e \in \gamma^+} \int_{\partial e} \left( \dot{\bar{t}} \cdot \delta u + (g^{\Delta u} n) \cdot n \bar{t} \cdot \delta u \right) ds - \sum_{\partial e \in \gamma^{c+}} \int_{\partial e} (\mathbf{grad}(\Delta u) + g^{\Delta u} n \otimes nI) \bar{t}^{c+} \cdot \delta u ds \\ & - \sum_{\partial e \in \gamma^{c-}} \int_{\partial e} (\mathbf{grad}(\Delta u) + g^{\Delta u} n \otimes nI) \bar{t}^{c-} \cdot \delta u ds, \end{aligned} \quad (38)$$

where we also add the contribution from the other side surface ( $\gamma^{c-}$ ) in the pair of the crack interface. The expressions for the paired crack surfaces  $\gamma^{c+}$  and  $\gamma^{c-}$  in the last two terms on the right-hand side in Equation (38) are similar to each other in terms of the integrands. However, they are independent and follow different deformation paths during traction release even though they share the same surface traction at the beginning.

**Remark 1.** In modeling crack opening or fracturing mechanics problems, the nodal force release algorithm is popular for reducing nodal forces on crack tails. Strictly speaking, the nodal force release method is only valid for small deformation problems. For finite deformation problems, it is difficult to derive a consistent formulation if nodal force algorithms are adopted. Our OTRA is elegantly described in terms of surface-related terminologies by following standard continuum mechanics theories. The consistent formulation on crack openings derived in this paper clearly indicates that crack surfaces also contribute to the tangent stiffness, which is crucial in achieving fast convergent rates for solving crack opening problems with finite formation.

**Remark 2.** Rather than directly releasing surface tractions, another option in DG formulations is to drop off the penalty term and replace the DG integral term in Equation (25) with an extrinsic cohesive law<sup>44,48</sup> (if data are available or can be artificially assumed) after crack openings are identified. However, this also implies a sudden switch on formulations and material laws during crack openings. Furthermore, to avoid infinite stiffness at the beginning of the softening curves in extrinsic CZM models, an artificial shift in the gap of cracks must be introduced. This switch on formulations and the artificial shift in the crack gaps may not induce serious instability issues for elastic systems but may cause instability issues for plasticity problems at large plastic deformation if implicit formulations are adopted. To improve the robustness of DG formulations for crack opening, we propose a DG-based relaxation scheme as follows.

### 5.3 | DG-based relaxation scheme

#### 5.3.1 | Motivation

As discussed in the University of Illinois Report on WARP3D<sup>54</sup> for crack propagation within CG frameworks, instability issues often occur using the nodal force release or the direct traction release method. Newly incorporated data structures for accommodating cracked surfaces in CG frameworks may be a potential source contributing to nonconvergent issues for implicit methods. However, we argue that there is an abrupt change in the behavior of two linearized systems, ie, for Problem A (the primary problem) without cracks and for Problem B (the auxiliary problem) with prescribed tractions on cracked surfaces where stresses are highly localized. Because of the implicit formulations adopted, the Newton-Raphson method is not robust in solving such a system with an abrupt change in behavior. New data structures for crack openings are avoided in DG frameworks. However, the direct traction release method adopted for crack openings in DG frameworks still inherits potential instability issues similar to CG methods. To improve the convergence for modeling crack opening problems, we propose a DG-based relaxation scheme that is also able to eventually enforce the traction-free condition on cracked surfaces. For an interior interface without cracking, the IIPG formulation in terms of the interior face integral in Equation (25) can be further expanded as follows:

$$\begin{aligned} R(u, \delta u) &= - \int_{\partial e} \{\sigma n\} \cdot [\delta u] ds + \delta_p \frac{G_0}{h_0} \int_{\partial e} [u] \cdot [\delta u] ds \\ &= - \int_{\partial e^+} \frac{\sigma^+ n^+}{2} \cdot \delta u^+ ds + \int_{\partial e^+} \frac{\sigma^+ n^+}{2} \cdot \delta u^- ds + \int_{\partial e^-} \frac{\sigma^- n^-}{2} \cdot \delta u^+ ds - \int_{\partial e^-} \frac{\sigma^- n^-}{2} \cdot \delta u^- ds \\ &\quad + \delta_p \frac{G_0}{h_0} \int_{\partial e^+} u^+ \cdot \delta u^+ ds - \delta_p \frac{G_0}{h_0} \int_{\partial e^+} u^+ \cdot \delta u^- ds - \delta_p \frac{G_0}{h_0} \int_{\partial e^-} u^- \cdot \delta u^+ ds + \delta_p \frac{G_0}{h_0} \int_{\partial e^-} u^- \cdot \delta u^- ds. \end{aligned} \quad (39)$$

If a crack opens on this interior face  $\partial e$ , the formulation for the direct traction release on the split two cracked surfaces  $\partial e = \partial e^+ \cup \partial e^-$  can be excerpted from Equation (38) as follows:

$$R_c(u, \delta u) = - \int_{\partial e^+} \tilde{t}^{c+} \cdot \delta u^+ ds - \int_{\partial e^-} \tilde{t}^{c-} \cdot \delta u^- ds. \quad (40)$$

We see that Equation (39) has a total of eight expanded terms of surface integrals and Equation (40) has only two terms of surface integrals. This implies that the two solution systems before cracking and after cracking are very different. An immediate switch between such two different systems may negatively impact the Newton-Raphson iterations. To avoid this abrupt change in solution systems, we propose a DG-based relaxation scheme as follows:

$$R(u, \delta u) = - \beta \int_{\partial e} \{\sigma n\} \cdot [\delta u] ds + \beta \delta_p \frac{G_0}{h_0} \int_{\partial e} [u] \cdot [\delta u] ds, \quad (41)$$

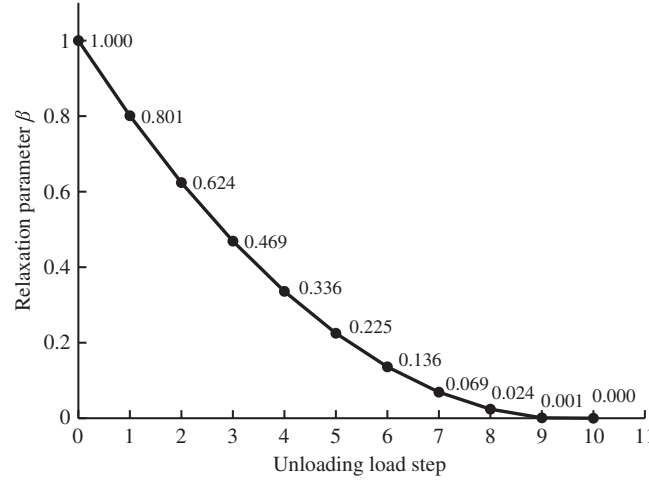
where only DG interface-related terms are included in Equation (41), and we simply introduce a relaxation parameter  $\beta$  before two surface integrals. If no crack opening occurs on this interior interface, we set up  $\beta = 1$ . If this interior face fails, we gradually reduce  $\beta$  to zero by following a substep unloading procedure. The traction-free condition for cracked surfaces will be completely recovered when  $\beta = 0$ . This is equivalent to CG methods where traction-free surfaces are not explicitly presented in formulations. As  $\beta$  is gradually changing, there will be a smooth transition between two systems without cracks and with cracks. The improvement in convergence for crack opening is, therefore, highly expected.

#### 5.3.2 | An adaptive relaxation scheme

Specifically, we have found an efficient way to gradually reduce the DG relaxation parameter by linearly lowering down the incremental of  $\Delta\beta$ . Our idea is to make larger reductions at the earlier stages of relaxation and smaller reductions at the final stages of relaxation. Designing such a relaxation scheme on  $\beta$  is motivated by the fact that the IIPG method allows a large range of the penalty parameter<sup>29,30,67</sup> and a substantial traction release actually occurs at  $\beta$  very close to zero. We have designed an incremental of  $\beta$  at substep  $k$  as follows:

$$\begin{cases} \Delta\beta^m = -\gamma; \\ \Delta\beta^k = \Delta\beta^{k-1} - h; & k = 1, 2, \dots, m \\ h = \frac{2(1-\gamma m)}{m(m-1)}, \end{cases} \quad (42)$$

where  $k$  is the  $k$ th substep,  $m$  is the total number of substeps in the relaxation process within a time/load step defined for interface openings,  $h$  is the constant, and  $\gamma$  is the incremental for the last relaxation substep. Practically,  $0.001 \leq \gamma \leq 0.01$



**FIGURE 5**  $\beta$  Reduction in 10 relaxation steps

and  $6 \leq m \leq 12$  may be recommended. The total relaxation of  $\beta$  at the  $k$ th relaxation substep is calculated as follows:

$$\beta^k = (m - k)\gamma h + \frac{(m - k)(m - k - 1)(1 - \gamma m)}{2m(m - 1)}. \quad (43)$$

Figure 5 plots the  $\beta$  reduction following relaxation steps for the case with  $m = 10$  and  $\gamma = 0.001$ .

*Remark 3.* It should be noted that the two predicted traction conditions of crack surfaces from the CZM method and the proposed DG relaxation method are different. There are two crack fronts in a CZM solution, ie, the physical crack front and the mathematical front. Indeed, the traction is free on the crack tail surfaces behind the physical front. However, an appreciable traction may still remain on the crack tail surfaces near the mathematical front even though a gap between cracked surfaces is visible and may take several additional time/load steps to reduce to zeros. Consequently, the crack length measured with respect to the mathematical front is longer than that measured using the physical front. We underline that the traction relaxation scheme proposed in Equation (43) targets the release of surface traction within only one (current) time/load step by the substepping technique without leaving traction residuals to the next time/load step. More precisely, a visible gap in crack tails from solutions using Equation (43) implies a unique front and a clear traction-free condition on cracked surfaces.

## 6 | NONLINEAR SOLUTION PROCEDURE

The final algebraic system equations for linearized *ProblemA* and *ProblemB* can be written as follows:

$$[\mathbf{K}]\delta\mathbf{U} = \delta\mathbf{R}, \quad (44)$$

where the stiffness  $\mathbf{K}$  and the residual  $\delta\mathbf{R}$  are derived from Equation (38) and Equation (31) by following standard finite element interpolation and integration procedures. Since the results of the algebraic system equations were documented in the work of Liu et al<sup>30</sup> for finite plasticity problems without cracks, we follow the same notations in the aforementioned work<sup>30</sup> and summarize the algebraic equations for finite plasticity problems with crack openings as follows:

$$\begin{aligned} [\mathbf{K}] = & \sum_{e \in \mathcal{E}} \int_e \mathbf{L}^T (\mathbf{C} + \Lambda^\sigma) \mathbf{L} dv - \beta \sum_{\partial e \in S/\alpha\gamma^c} \int_{\partial e} [\mathbf{N}^T] \mathbf{n} \{ (\mathbf{C} + \Lambda^\sigma) \mathbf{L} \} d\mathbf{s} \\ & + \beta \sum_{\partial e \in S/\alpha\gamma^c} \frac{\delta_p G_0}{h_0} \int_{\partial e} ([\mathbf{N}^T][\mathbf{N}] + \mathbf{N}^T \Lambda^{[u]} \mathbf{n}^T \mathbf{n} \{ \mathbf{L}_0 - \mathbf{L}_T \}) d\mathbf{s} - \sum_{\partial e \in \gamma^i} \int_{\partial e} \mathbf{N}^T \left( \Phi^i + \Lambda^i \mathbf{n}^T \mathbf{n} (\mathbf{L}_0 - \mathbf{L}_T) \right) d\mathbf{s} \\ & - \alpha \sum_{\partial e \in \gamma^{c+}} \int_{\partial e} \mathbf{N}^T \Lambda^{\bar{i}c+} (\mathbf{L} + \mathbf{n}^T \mathbf{n} (\mathbf{L}_0 - \mathbf{L}_T)) d\mathbf{s} - \alpha \sum_{\partial e \in \gamma^{c-}} \int_{\partial e} \mathbf{N}^T \Lambda^{\bar{i}c-} (\mathbf{L} + \mathbf{n}^T \mathbf{n} (\mathbf{L}_0 - \mathbf{L}_T)) d\mathbf{s} \end{aligned} \quad (45)$$

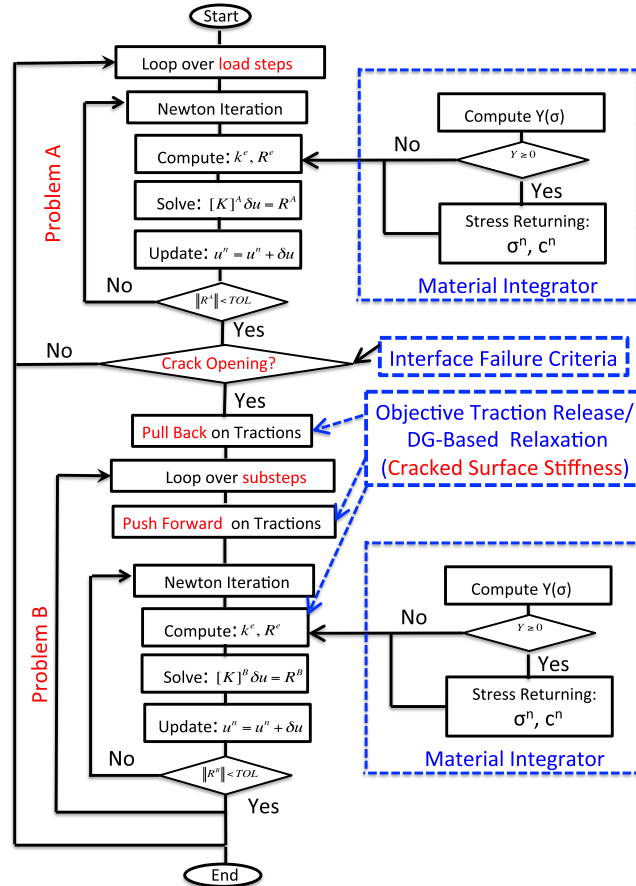
and

$$\begin{aligned} \mathbf{R} = & \sum_{e \in \chi} \int_e \mathbf{L}^T \boldsymbol{\sigma} d\mathbf{v} - \sum_{\partial e \in S/\alpha\gamma^c} \int_{\partial e} [\mathbf{N}^T] \mathbf{n} \{ \boldsymbol{\sigma} \} d\mathbf{s} + \beta \sum_{\partial e \in S/\alpha\gamma^c} \frac{\delta_p G_0}{h_0} \int_{\partial e} [\mathbf{N}^T] [\mathbf{u}] d\mathbf{s} - \sum_{e \in \chi} \int_e \mathbf{N}^T \mathbf{b} d\mathbf{v} \\ & - \sum_{\partial e \in \gamma^l} \int_{\partial e} \mathbf{N}^T \bar{\mathbf{t}} d\mathbf{s} - \alpha \sum_{\partial e \in \gamma^{c+}} \int_{\partial e} \mathbf{N}^T \bar{\mathbf{t}}^{c+} d\mathbf{s} - \alpha \sum_{\partial e \in \gamma^{c-}} \int_{\partial e} \mathbf{N}^T \bar{\mathbf{t}}^{c-} d\mathbf{s}, \end{aligned} \quad (46)$$

where  $\Lambda^\sigma$ ,  $\Lambda^{[u]}$ ,  $\Lambda^{\bar{\mathbf{t}}}$ ,  $\Lambda^{\bar{\mathbf{t}}^{c+}}$ ,  $\Lambda^{\bar{\mathbf{t}}^{c-}}$ , and  $\mathbf{n}$  are matrices in terms of stress, jump displacement, traction on  $\gamma^l$ , traction on crack surface  $\gamma^{c+}$ , traction on crack surface  $\gamma^{c-}$ , and surface normal directional matrix, respectively. *ProblemA* and *ProblemB* are implemented through setting the parameter  $\alpha$  in Equations (45) and (46) to  $\alpha = 0$  and  $\alpha = 1$ , respectively. Moreover, for the DG-based relaxation scheme,  $\beta$  will be gradually relaxed to zero for  $\partial e \in S^c$  while  $\alpha = 0$ .  $\beta$  is always equal to 1 for  $\partial e \in S^l$ . For the interpolation matrices in Equation (45) and Equation (46), we reproduce them from the work of Liu et al<sup>30</sup> as follows:

$$\begin{cases} \Delta u = \mathbf{N} \Delta \mathbf{U}; \\ \mathbf{grad} \Delta u = \mathbf{L} \Delta \mathbf{U}; \\ \mathbf{grad}^T \Delta u = \mathbf{L}_T \Delta \mathbf{U}; \\ \mathbf{trace}(\mathbf{grad} \Delta u) \mathbf{I} = \mathbf{L}_0 \Delta \mathbf{U}; \\ g^{\Delta u} = (\mathbf{L}_0 - \mathbf{L}_T) \Delta \mathbf{U}; \\ \Psi^{\Delta u} = \Psi^{\bar{\mathbf{t}}} \Delta \mathbf{U}. \end{cases} \quad (47)$$

In Equation (45),  $C$  is a matrix form for the algebraic tangent material modulus. To achieve a fast convergent rate for plasticity problems,  $C$  must be obtained from solving a local material integration problem in which stress return algorithms should be consistently implemented. As we focus on crack openings in this work, we refer to the work of Liu et al<sup>30</sup> for the detailed derivation on local material integrators.



**FIGURE 6** diagram of nonlinear solution procedures for solving crack openings in finite plasticity media. DG, Discontinuous Galerkin [Colour figure can be viewed at wileyonlinelibrary.com]

In Figure 6, we present a detailed diagram on the IIPG solution procedures for solving crack opening problems. This diagram serves as an adjointer between the formulation and the implementation of the proposed IIPG method. The IIPG finite element program starts to loop over load steps. In each load step, *ProblemA* is first solved for finite plasticity without crack openings. Particularly, element stiffness matrices and residual forces as well as DG interface stiffness and residual forces are calculated. Stress returning and algebraic tangent modulus are provided in the material integrator module. After *ProblemA* converges for nonlinear iterations, crack openings are checked according to the converged solutions of *ProblemA*. If there are crack openings, *ProblemB* is solved within the current load step by applying the direct traction release or the DG-based relaxation scheme to release tractions on crack surfaces. The OTRA is followed to enforce the direct traction release scheme. Before starting to release traction, traction on cracked surfaces is first pulled back to the reference configuration. These pulled-back tractions are fixed in their directions in the reference configuration and are pushed forward into the current configuration to provide new tractions to be released in the spatial domain. In order to achieve converged solutions in *ProblemB*, substep algorithms must be performed. In each substep, the same nonlinear iteration and stress returning procedures as *ProblemA* are followed. After traction on cracked surfaces is successfully released, *ProblemB* is finished, and the solution is going to the next load step. It should be mentioned that some automatic time-stepping algorithms and automatic substep algorithms for traction release are also implemented in our IIPG code, which are necessary for modeling complex crack opening problems.

## 7 | NUMERICAL EXAMPLES

In this section, we present two numerical examples to demonstrate the capability and performance of the proposed implicit DG computational framework in modeling ductile metal fracture problems using both hyperelastoplastic and hypoelastoplastic models. In these two examples, mode I-dominated crack openings are focused on. The first example is to model the multicrack failure of a nuclear fuel rod due to the formation of hydrogen-induced hydrides where the burst failure mode is simulated. The modeling of fuel rod failure problems is very important to nuclear reactor design and its safety analysis. Currently, it is a great challenge in achieving a complete failure simulation for such structures undergoing large plastic deformation. This problem is solved twice by our proposed IIPG method along with both the traditional direct traction release and the DG-based relaxation method. The performance of the DG formulation along with the traction relaxation scheme is further evaluated in solving the Sandia Fracture Challenge,<sup>63</sup> which is a more complex ductile failure problem that has become a popular benchmark in the fracture mechanics community. Furthermore, we mention that a good prediction of our DG code on the first crack opening in the second example when compared to the experimental displacement-load curves may serve as a partial validation on the code.

To solve these two problems, we further assume that a mode I-dominated crack opening should satisfy the following three conditions on DG interfaces simultaneously:

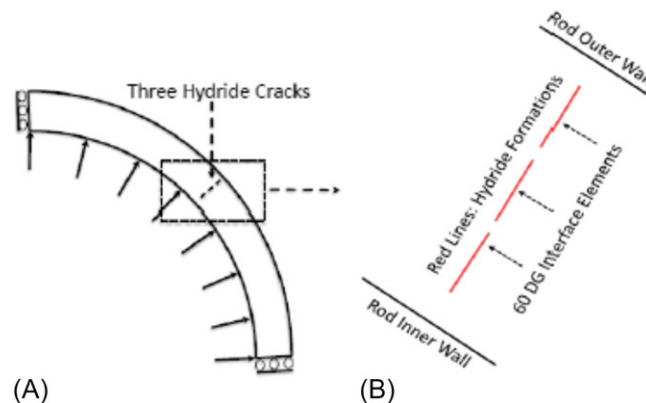
$$\left\{ \begin{array}{l} \bar{\epsilon}^p \geq \bar{\epsilon}^F; \\ T_n = (\sigma n) \cdot n > 0; \text{ and} \\ Y(\sigma, \sigma_0) = 0, \end{array} \right. \quad (48)$$

where  $\bar{\epsilon}^F$  is the failure strain and  $T_n$  is the normal stress on DG interfaces. More precisely, for a crack opening on a DG interface, the equivalent plastic strain should reach a failure limit, the normal stress should be positive (in tension), and materials should be in the yield state and not in the elastic unloading condition. The condition of the normal stress in tension in Equation (48) is necessary to prevent unexpected crack openings in compression zones where equivalent plastic strains may also reach strain failure limits. Furthermore, with regard to complex loading and unloading stress states for some areas resulting from crack openings in other areas, the checking on the yield condition specified in Equation (48) is to avoid unrealistic interface openings where the equivalent plastic strain reaches the failure limit but materials are in the elastic state.

## 7.1 | Nuclear fuel rod failure

### 7.1.1 | Model description

Figure 7 presents an illustration of a quarter plane strain model for a fuel rod burst failure due to the expansion of fuel material under high temperature. The geometry data for the rod are obtained from the work of Nilsson et al.<sup>75</sup> The outer radius of the rod is 5.0305 mm, and the inner radius is 4.4165 mm. The wall thickness is 0.614 mm. The rod is assumed to be made of a typical Zircaloy-4 cladding material. The material model is the neo-Hookean hyperelastic model with  $J_2$  plasticity. The two Lamé constants corresponding to the bulk and shear moduli for the small linear elasticity theory, the yield strength of Zircaloy-4, and Voce's isotropic hardening function based on the uniaxial experiments for Zircaloy materials in the work of Nilsson et al.<sup>75</sup> are listed in Table 1. Finally, we define the interface parameters. As hydride formations are typically very thin in one direction and very long in the other direction, especially for reoriented hydride formations, we model hydride formations as interfaces without thickness. More precisely, the hydride bulk matrix itself is not explicitly modeled. The hydride/bulk matrix interface failure strength in Equation (10) is assumed to be 300 MPa, roughly 85% of the bulk matrix yield strength. The shear contribution coefficient for interface debonding in Equation (10) is set to 0. We set up three hydrides with equal length and distributed along a straight line in the radial direction. The length of each hydride is 0.1228 mm. The thickness between hydride tips is 0.0614 mm. Three cracks will eventually initiate at the locations of these three hydrides by the debonding of the hydride/bulk matrix. The finite element model contains 2000 elements and 100 DG interfaces. A fine mesh is designed near hydride interface areas. We apply radial displacement on the inner wall up to 2 mm to mimic the thermal expansion loading acting on the tube inner wall surface. Besides the performance evaluation on the proposed DG formulation, we are also interested in how hydride interfaces debond to form cracks and how cracks grow and coalesce.



**FIGURE 7** A quarter plane strain discontinuous Galerkin (DG) interface model for zircaloy cladding rods. A, illustration of geometry and cracks; B, A total of 60 dg interface elements for hydride formations [Colour figure can be viewed at [wileyonlinelibrary.com](http://wileyonlinelibrary.com)]

**TABLE 1** Summary of material model and fracture failure parameters

| Material Parameter      |                                                                                    |
|-------------------------|------------------------------------------------------------------------------------|
| Hyperelastic model      | neo-Hookean                                                                        |
| Bulk modulus            | 95 MPa                                                                             |
| Shear modulus           | 57 MPa                                                                             |
| Yield strength          | 339 MPa                                                                            |
| Hardening function      | $\bar{\sigma}_0 = 339 + 887\bar{\epsilon}^p + 101(1 - e^{-151.8\bar{\epsilon}^p})$ |
| Failure model           | Johnson-Cook: $\epsilon^f = D_1 + D_2 e^{-D_3 \sigma^*}$                           |
| $D_1, D_2$ , and $D_3$  | 0.4, 0.0, 0.0 MPa <sup>-1</sup>                                                    |
| Hydride debonding model | $\sigma_n + \gamma \tau_s > \sigma_0$                                              |
| $\gamma$ and $\sigma_0$ | 0.0, 300 MPa                                                                       |

**TABLE 2** Nonlinear iteration and  $L^2$  norm of residual force: scheme 1, direct traction release; scheme 2, discontinuous Galerkin relaxation

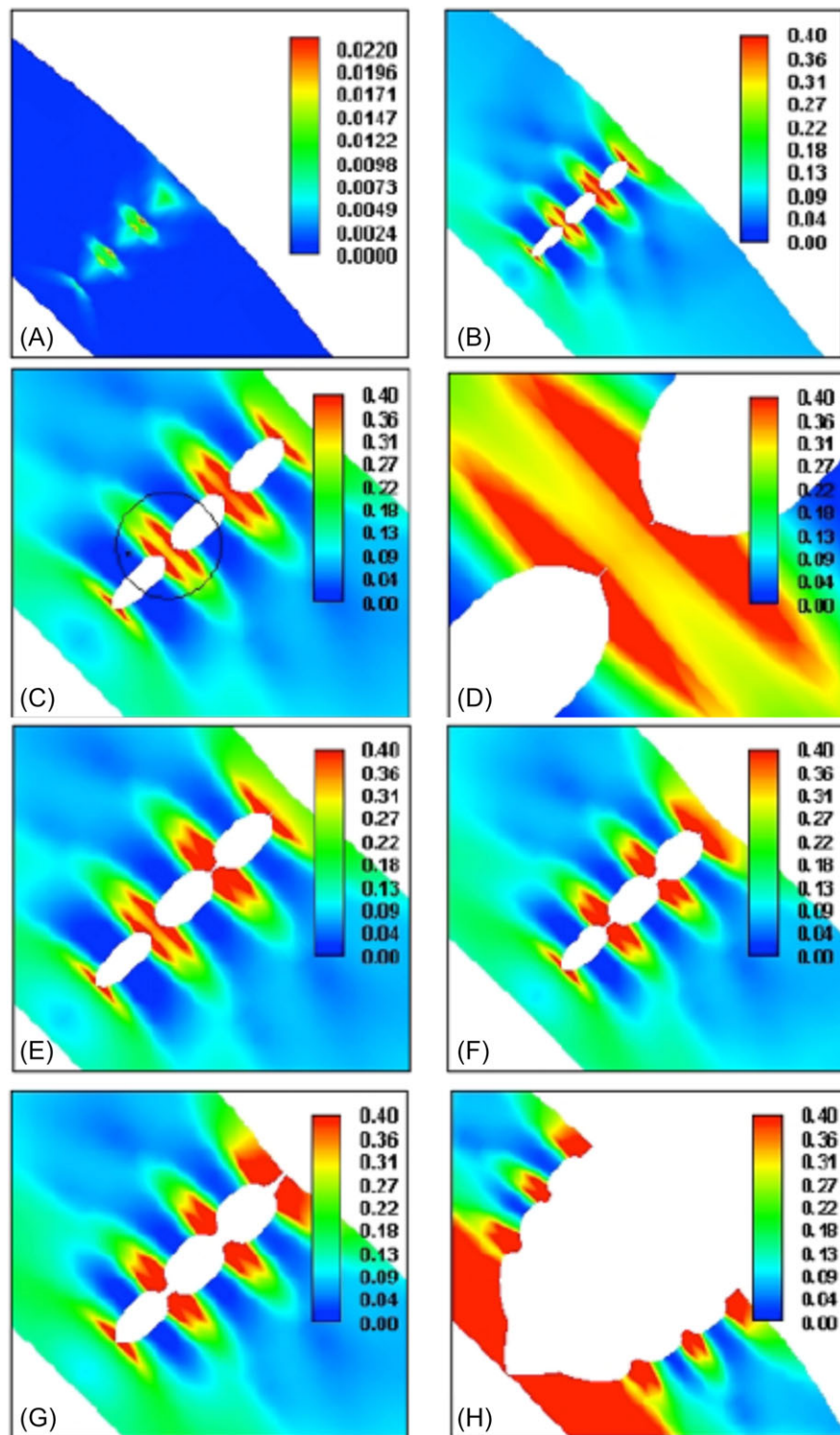
| Iteration | Elasticity | Plasticity | Interface Debonding | Ligament Breakage |
|-----------|------------|------------|---------------------|-------------------|
| 1         | 853.39     | 1713       | 10489               | 15612             |
| 2         | 4.19e−5    | 0.116      | 4.09                | 4.69              |
| 3         | (Scheme 1) | 2.49e−2    | 0.114               | 0.34              |
| 4         | (Scheme 2) | 1.52e−7    | 1.0e−3              | 1.87e−3           |
| 5         |            | (Scheme 1) | 1.29e−6             | 2.95e−5           |
| 6         |            | (Scheme 2) | (Scheme 1)          | (Problem A)       |
| 7         |            |            | (Scheme 2)          |                   |
| 8         |            |            |                     | 0.162             |
| 9         |            |            |                     | 8.75e−2           |
| 10        |            |            |                     | 1.42e−2           |
| 11        |            |            |                     | 9.56e−4           |
| 12        |            |            |                     | (Scheme 2)        |

### 7.1.2 | Performance

Table 2 presents the convergent patterns of Newton-Raphson iterations in solving Problem A and Problem B using the IIPG method. Again, Problem A is to solve the problem without crack opening (it may include opened interior faces where traction is already released). As discussed earlier, Problem B is to release tractions on cracked surfaces by taking the solution from Problem A as its initial condition. Two schemes, namely, direct traction release and DG-based relaxation, are adopted and tested along with our IIPG framework for this rod failure problem. The number of substeps for solving Problem B for both the direct traction release and DG-based relaxation schemes is set up by 5. In the Table, scheme 1 represents the direct traction release method, and scheme 2 refers to the DG-based relaxation method. The residual tolerance for convergence is set up by  $1.0e-3$ . The initial total load steps are 2000. At the early loading stages, materials are in elastic states but still nonlinear due to the hyperelasticity model, and only two iterations are needed to achieve a converged solution for each load step. Following the increase in loading, materials yield and four Newton-Raphson iterations are needed to reach convergence. In the plastic stages, cutback operations on loading steps are necessary for achieving converged solutions. Solutions at interface debonding and crack initiation need four or five iterations. Quadratic convergent rates are observed in the stages of crack initiation with plasticity. For the direct traction release method, solutions converge only at the crack initiation stages where interface failure is judged by the local stress criteria and only 2%-3% equivalent plastic strain occurs. However, it does not converge at the ligament tearing-off stages where crack opening is evaluated by the plastic strain up to 40%. We conclude that the direct traction release method has instability issues for crack opening in media with complex and large plastic strain. On the other hand, this rod failure problem is successfully solved by the IIPG method along with the proposed DG-based relaxation scheme. The first part in the last column in Table 2 lists a typical convergent pattern for solving Problem A at the ligament tearing-off stages. The second part in the last column in the Table corresponds to Problem B. As shown in the last column in Table 2, quadratic convergent rates or at least fast convergent rates are observed in the DG relaxation method for crack openings at the ligament tearing-off stages.

### 7.1.3 | Simulation result and discussion

We now discuss the simulation results obtained from our IIPG formulation. The discussion is based on the solution obtained from the DG-based relaxation scheme as the direct traction release scheme does not converge for crack opening at large plastic strain for this rod problem. It should be noted that hydrides themselves are initially not cracks, and they are treated as particles or inclusions from the micromechanics point of view. More precisely, the material behaviors of these hydrides are substantially different from their surrounding bulk metal matrix. Separation of hydrides from the bulk matrix eventually occurs through hydride/bulk matrix interface debonding. Figure 8A presents a contour for crack initiation at displacement loading  $U = 0.019$  mm. A total of 60 hydride/bulk matrix interface elements completely debond, and three cracks are initiated. The equivalent plastic strain is contoured in Figure 8A-H. As expected, from Figure 8A, relatively high plastic strain occurs at crack tips. Following the increase in loading, the three initiated cracks start to grow, and the corresponding three voids are formed due to the development of substantial plastic strain in the ductile bulk matrix. These three voids are plotted in Figure 8B,C at displacement loading  $U = 0.501$  mm. A detailed large-plastic-strain profile at the



**FIGURE 8** Incomplete interior penalty galerkin simulation on nuclear fuel rod failure due to fuel thermal expansion. A, three hydride cracks are initiated through debonding at  $U = 0.019$  mm (Traction release and Discontinuous Galerkin [DG]-based relaxation methods); B, three voids grow and substantial plastic strain develops before ligaments break at  $U = 0.501$  mm (Traction release and DG-based relaxation methods); C, ligaments between voids start to shrink and break at  $U = 0.511$  mm (no convergence is observed for the traction release method; (C) and (D) contours are results from the dg-based relaxation method); D, ligaments between the first void and the second void start to break at  $U = 0.511$  mm (zoomed-in part from the circle area indicated in (C)); E, ligaments between the second void and the third void completely tear off at  $U = 0.521$  mm; F, ligaments between the first void and the second void completely tear off at  $U = 0.581$  mm; G, ligaments between the third void and the rod exterior wall completely tear off at  $U = 0.604$  mm; H, ligaments between the first void and the rod interior wall completely tear off at  $U = 1.71$  mm [Colour figure can be viewed at [wileyonlinelibrary.com](http://wileyonlinelibrary.com)]

area between the first void and the second void is zoomed in and plotted in Figure 8D. On the other hand, the areas between voids represented by ligaments start to shrink in their thicknesses. This phenomenon is called ligament “necking” in terms of micromechanics terminologies. The necking of ligaments continues following the increase in loading and eventually results in ligament breaking or tearing-off due to the large plastic strain developed. Ligament breaking results in the coalescence of voids, which implies a significant reduction in structure load capacity. The coalescence of the second void and the third void is presented in Figure 8E. All three voids eventually coalesce together into a large void at displacement loading  $U = 0.581$  mm, which is shown in Figure 8F. Following the coalescence of interior voids, the ligaments in the area near the exterior wall of the rod tear off at  $U = 0.604$  mm. This process can be viewed in Figure 8G. As shown in Figure 8H, a complete rod structure failure occurs at  $U = 1.71$  mm following the necking and tearing-off of the ligaments in the interior wall of the rod.

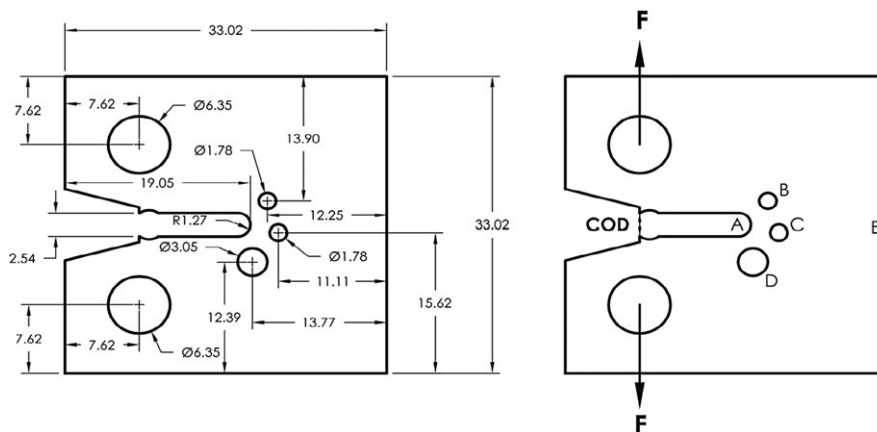
## 7.2 | The Sandia Fracture Challenge

### 7.2.1 | Model summary

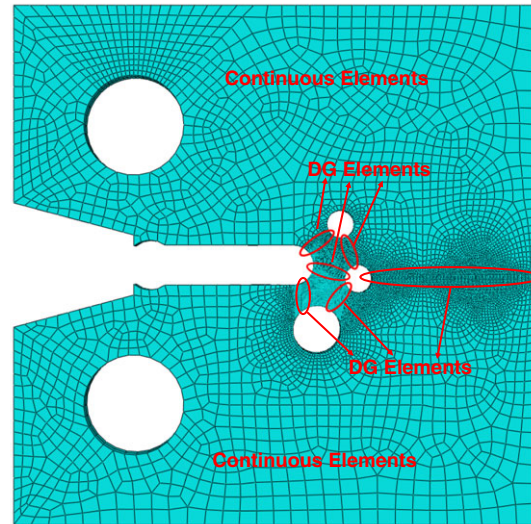
For the second example, we use the Sandia Fracture Challenge. In this work, we focus on the performance evaluation of the proposed numerical formulations. For modeling the Sandia Fracture Challenge using various material models, we refer to the works of Gross and Ravi-Chandar,<sup>60</sup> Bao and Wierzbicki,<sup>62</sup> Boyce et al.,<sup>63</sup> and Xue.<sup>73</sup> The traditional hypoe-elastoplasticity model is used, and the the Johnson-Cook material failure criterion is adopted. The parameters of the hardening function and the failure model were calibrated using the experimental data of the bar tests reported in the work of Boyce et al.<sup>63</sup> All material model parameters are summarized in Table 3. The geometry of this problem is presented in Figure 9, where a total of three holes were drilled to represent defects and two additional large holes were designed for applying loadings. Figure 10 presents a relatively fine finite element mesh for the Sandia Fracture Challenge. A half model is adopted due to symmetry with regard to the thickness direction. In this mesh, two hexahedral elements are meshed along the thickness direction. Moreover, 60 elements are used along each section of A-C, A-B, A-D, B-C, and C-D (see Figure 9), and 100 elements are distributed along the C-E section. A total of 37 372 hexahedral elements are meshed in the model, and more specifically, a total of 4272 DG interface elements are applied by breaking a total of 1432

**TABLE 3** Summary of material model parameters

| Material Parameter     |                                                                       |
|------------------------|-----------------------------------------------------------------------|
| Young's modulus        | 199 000 MPa                                                           |
| Poisson ratio          | 0.30                                                                  |
| Hardening function     | $\bar{\sigma}_0 = 755.4 + 818.1(0.05674 + \bar{\epsilon}^p)^{0.2889}$ |
| Plastic model          | $J_2$                                                                 |
| Yield strength         | 1112.5 MPa                                                            |
| Failure model          | Johnson-Cook: $\epsilon^f = D_1 + D_2 e^{-D_3 \sigma^*}$              |
| $D_1, D_2$ , and $D_3$ | 0.4, 3.810, 1.847 MPa <sup>-1</sup>                                   |



**FIGURE 9** Dimensions of the Fracture Challenge Specimen Geometry in Millimeters. The Plate Thickness is 3.124 mm.<sup>63</sup> COD, Crack Opening Displacement (Reprinted With Permission)

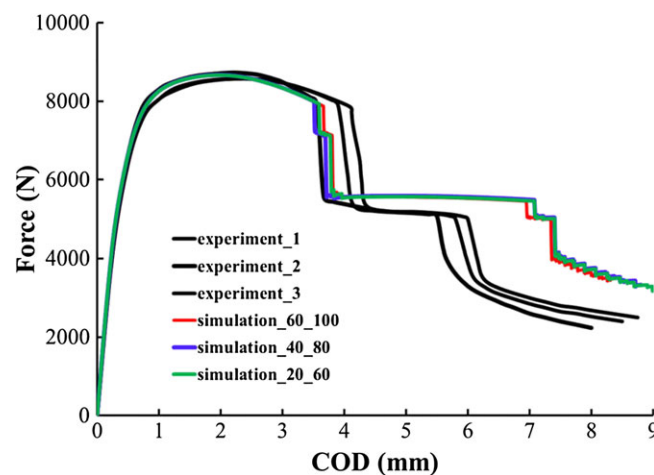


**FIGURE 10** Mesh of the Sandia Fracture Challenge Model. DG, Discontinuous Galerkin [Colour figure can be viewed at [wileyonlinelibrary.com](http://wileyonlinelibrary.com)]

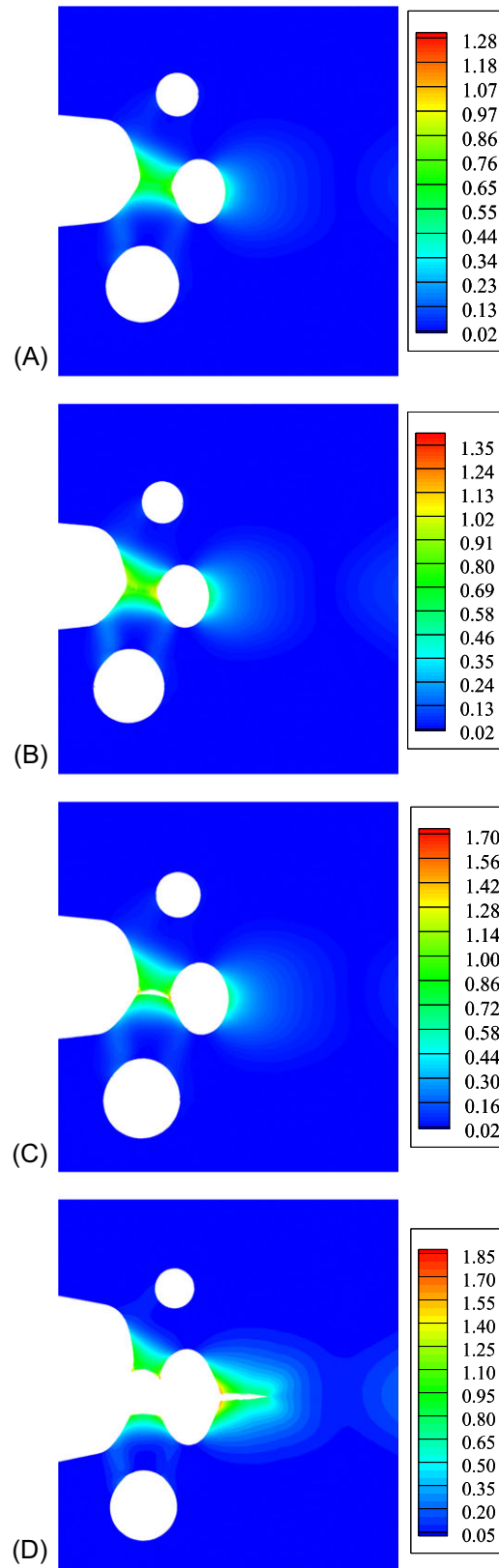
continuous elements in the areas with high stress concentration, as illustrated in Figure 10. A majority of elements are continuous elements. The residual tolerance for convergence is set up by  $1.0e - 3$ . To capture a detailed crack opening behavior, we set up 8000 time steps for crack opening displacement (COD) loading and 12 substeps for DG relaxation for each crack opening.

### 7.2.2 | Prediction and performance

We now discuss the simulation results from our IIPG method and compare them with the Sandia experiments. Figure 11 plots the force-COD curve from our simulation and the Sandia experiments corresponding to the A-C-E mode. Several snapshots for predicted fracture initiation and propagation as well as the evolution of the equivalent plastic strain for the Sandia benchmark are presented in Figure 12. From Figure 11, our predicted reaction force at COD = 1 mm is 8258 N, and the corresponding reaction forces from experiments are between 8000 and 8300 N. The peak reaction force predicted from the IIPG method is 8672 N, and the experimental results are between 8500 and 8800 N. The predicted crack initialization occurs at COD = 3.66 mm and force = 7974 N, whereas in the experiments, the COD at crack initiation ranges between 3.4 and 4.0 mm, and force ranges between 8000 and 8200 N. After the first crack (A-C) is open, the predicted force drops to 5528 at COD = 3.95 mm. The corresponding force from experiments drops to between 5400 and 5500 N at COD = 3.50–4.1 mm. In summary, our DG prediction for the onset of the first crack (A-C) and the dropping of the reaction



**FIGURE 11** Crack opening displacement (COD) versus load. Comparison of Discontinuous Galerkin relaxation prediction with experiments excerpted from the work of Boyce et al<sup>63</sup> for the A-C-E mode only (Reprinted with permission) [Colour figure can be viewed at [wileyonlinelibrary.com](http://wileyonlinelibrary.com)]



**FIGURE 12** Equivalent plastic strain contour of the Sandia Fracture Challenge Model. A, First crack initialization at COD = 3.343 mm; B, First crack propagation at COD = 3.579 mm; C, First crack open at COD = 3.845 mm; D, second crack at COD = 6.396 mm. COD, Crack opening displacement [Colour figure can be viewed at [wileyonlinelibrary.com](http://wileyonlinelibrary.com)]

**TABLE 4** Global nonlinear iterations and  $L^2$  norm of residuals for the Sandia Fracture Challenge

| Iteration | $L^2$ Norm of Residuals |            |               |
|-----------|-------------------------|------------|---------------|
|           | Elasticity              | Plasticity | Crack Opening |
| 1         | 9.88+00                 | 7.73+00    | 7.83          |
| 2         | 2.11-04                 | 4.94-02    | 4.46-01       |
| 3         |                         | 6.05-04    | 1.27e-02      |
| 4         |                         |            | 2.83e-04      |

force after the first crack opening is very good. The predicted second crack occurs at COD = 6.94 mm, and the predicted force drops to 3960 N. The average experimental COD for the second crack is 5.75 mm, and the force drops to 3800 N. After the first crack opens, there is a broad plateau with a length up to 2 mm observed from the experiments. Our simulation does show a plateau starting at COD = 3.95 mm, but the predicted plateau length before the second crack opening (C-E) is longer than that from the experiments. Similar longer plateaus and delayed crack openings for the second crack have also been reported using other numerical methods as in the works of Gross and Ravi-Chandar<sup>60</sup> and Boyce et al,<sup>63</sup> where this large discrepancy between numerical predictions and experimental results with regard to the onset of the second crack is discussed in detail. This discussion in the work of Boyce et al<sup>63</sup> suggests that more appropriate material failure models should be calibrated and developed using more test data to improve the prediction. To study the effect of different meshes to the predicted COD-force curve, two additional coarse meshes are designed in the high-stress-concentration areas near holes where DG elements are applied. In the second mesh, 40 elements are used along each section of A-C, A-B, A-D, B-C, and C-D areas, and 80 elements are distributed along the C-E section. A total of 22 622 hexahedral elements are meshed. For the third mesh, 20 elements are used along each section of A-C, A-B, A-D, B-C, and C-D areas, and 60 elements are distributed along the C-E section. A total of 14 130 hexahedral elements are meshed. The predicted three COD-force curves corresponding to these three different meshes are plotted and compared in Figure 11. We conclude that the predicted solutions from these three different meshes are very similar, including the onset of the crack opening, the dropping of the reaction forces, and the length of the plateaus.

We now discuss the performance of the IIPG framework. It should be mentioned that an automatic cutback and increase scheme in time/load steps was implemented in the code. This scheme is crucial in achieving the complete simulation for this complex Sandia Fracture Challenge problem. Table 4 presents profiles of  $L^2$  residual reduction for global nonlinear iterations for our IIPG solver in solving this benchmark problem. The Table shows typical convergent patterns at different deformation stages, including elastic, plastic, and crack opening stages. We conclude that while exact quadratic convergent rates are not often observed in ductile tearing stages, convergent rates are still fast, and solutions are typically finished in three to five iterations. In summary, we see that crack initiation, growth, coalescence, and tearing are extremely complex and that problems are highly nonlinear and very difficult to simulate. It is concluded that our IIPG formulation along with the DG-based relaxation scheme demonstrates great potential for modeling multiple crack openings in finite plasticity media. For problems with a large amount of cracks, there are potential serious mesh distortions around inclusions at high load levels, which may have negative impacts on the convergence of solutions. The use of orthogonal meshes in the potential ligament failure areas as discussed in the works of Liu et al<sup>30</sup> and Moran et al<sup>76</sup> and arbitrary Lagrangian-Eulerian finite elements<sup>77</sup> in the IIPG framework is the proposed future research direction by the authors for problems with more complex multiple cracks.

## 8 | CONCLUSIONS

An implicit DG framework based on the IIPG formulation has been proposed to model crack openings in ductile materials undergoing finite plastic deformation. To achieve fast convergent rates, our IIPG formulation has been consistently linearized by taking into account the geometrical stiffness contribution from cracked surfaces. An objective traction release scheme within the framework of the direct traction release and an adaptive DG-based relaxation method have been implemented and tested with both hyperelastoplastic and hypoelastoplastic frameworks. The DG-based relaxation method is much more robust than the direct traction release to obtain complete simulations on crack openings in media especially undergoing large plastic deformation. We have demonstrated a good performance of the IIPG formulation along with the DG-based relaxation scheme by solving a nuclear fuel rod failure problem with multiple hydride crack openings and the Sandia Fracture Challenge.

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